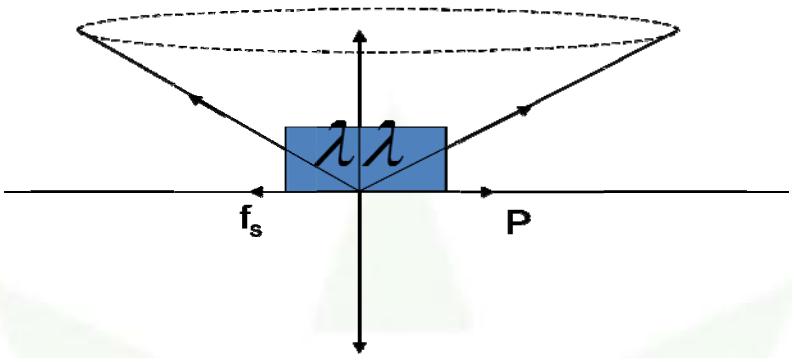
Angle of Repose (Ө): Angle of inclination of the plane with the block just starts sliding is known as angle of repose.

Angle of friction or cone of friction:



Given: W = weight of the block

P = force applied on the block for which the block just sliding

The other force acting on the block is

(1) The reaction of the table on the block R perpendicular to the plane along upward direction.

(2) The maximum value of static frictional force along opposite to the applied force.

Since the block is in equilibrium.

Let F be the resultant of the R and f_s. Let F be the resultant of R and P which makes an angle λ with the vertical.

λ is known as the angle of friction.

From the figure

$$\tan\lambda = \frac{f_s}{R} = \mu_s$$
$$\mu_s = \tan\lambda$$

We have seen earlier that $\mu_s = \tan\theta$ when Ө is the angle of repose.

$$\therefore \tan\lambda = \tan\theta$$
$$or\ \lambda = \theta$$

Angle of friction = Angle of repose